

Random Processes Quiz

Q. One: For the process $X(t) = 6\cos(t + \theta)$ where $\theta \sim U(0, \pi)$, we have $E[X^2(t)] =$

- A) 0.5π B) π ✓ C) 2π D) 3π E) None

Q. Two: The variance of a mean-ergodic process $X(t)$ with no periodic component and $R_{xx}(\tau) = \pi + \pi^2 e^{-|\tau|}$ is equal to

- ✓ A) π^2 B) π C) 2π D) 3π E) None

Q. Three: For a random process $X(t) = 3A\cos(t)$ where $A \sim U(0, \pi)$ we have $P_{xx} =$

- A) $\frac{1}{12}$ B) $\frac{1}{2}$ C) $\frac{1}{3}$ D) $\frac{1}{6}$ ✓ E) None

Q. Four: For $X \sim \text{Poisson}(m)$ with $P(X = 0) = \sqrt{\frac{1}{e}}$; then $P(X > \bar{X}) =$

- A) $\sqrt{\frac{1}{e}}$ ✓ B) $\frac{1}{e}$ C) $1 - \sqrt{\frac{1}{e}}$ D) $1 - \frac{1}{e}$ E) None E) None

Q. Five: Jointly wide-sense stationary orthogonal processes $X(t), Y(t)$ with $R_{xy}(\tau) = 1 + \bar{X}\bar{Y}$, then $C_{xy}(t, t + \tau) =$

- A) 1 ✓ B) 0 C) 2 D) 3 E) None

Q. Six: For jointly wide-sense stationary uncorrelated discrete time random processes $X(nT_s), Y(nT_s)$ such that $C_{xy}(nT_s, nT_s + kT_s) = 4 - \bar{X}$ then $R_{xy}(kT_s) + \bar{X}\bar{Y} =$

- A) 2 ✓ B) 4 C) 6 D) 8 E) None

Q. Seven: If $\text{Var}(X) = 3$, and $Y = X - \bar{X}$, then the correlation of X and Y is

- A) 1 ✓ B) 2 C) 3 D) 4 E) None

Q. Eight: The pdf of $Y = 4X + 1$; where $X \sim U(0, 1)$, equals

- A) 2 ✓ B) 0.25 C) 0.5 D) 1 E) None

Q. Nine: The variance of the process $X(t) = A$ where $A \sim U(-\pi, \pi)$ is equal to

- A) $\frac{2\pi^3}{3}$ B) $\frac{\pi^3}{3}$ C) $\frac{\pi^2}{3}$ ✓ D) $\frac{2\pi^2}{3}$ E) None

Q. Ten: For $X \sim N(a, a)$ if $\text{std}(X) = E(X)$, then $a + \text{std}(X) =$

- A) 1 B) 4 C) 6 ✓ D) 2 E) None

Q. Eleven: If $\alpha > 0$ be a number that satisfies $(-1)^\alpha = -1$. If we have $E(X^2(t)) = t^\alpha + \pi + \sin(t)$ then P_{XX}

- A) π ✓ B) 3π C) 2π D) 0 E) None

Q. Twelve: Which of the following is false?

✓ A) The null hypothesis is the one to be tested.

B) Null and alternative hypotheses cannot be both true at the same time.

C) For $X \sim N(1,2)$ then $P(X > 2) = 0.5$.

D) Type I error occurs when we reject H_0 when it is actually true.

Q. Thirteen: Consider the joint pdf $f(X, Y) = x$ when $0 < x < 2, 0 < y < x$ and zero otherwise. Then $f_Y(y) = \frac{1}{2}(4 - y^2)$.

- A) $0.5x$ B) X C) $2y$ ✓ D) 2 E) None

Q. Fourteen (✓, X): For any constant $c > 0$, we have $E(cX) = cE(X)$.

✓

Q. Fifteen (✓, X): A correlation coefficient of 0.97 indicates a strong positive correlation.

✓

Q. Sixteen (✓, X): A strictly stationary process is always wide-sense stationary.

✓

Q. Seventeen (✓, X): For independent random variables X and Y , $E(XY) = E(X)E(Y)$.

✓

Q. Eighteen (✓, X): In a binomial distribution $X \sim B(n, p)$, the variance is $np(1 - p)$.

✓

Q. Nineteen (✓, X): For $X \sim N(\mu, \sigma^2)$, the random variable $Z = \frac{X - \mu}{\sigma}$ follows a standard normal distribution $N(0, 1)$.

Q. Twenty (✓, X): The sample mean is an unbiased estimator of the population mean.

✓